ROUND ROBIN

```
#include <stdio.h>
int main() {
    int n, tq;
    int bt[10], rt[10], wt[10], tat[10];
    int time = 0, completed = 0;
    float avg_wt = 0, avg_tat = 0;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter burst time of each process:\n");
    for (int i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
        rt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &tq);

    printf("\nGantt Chart:\n");
    printf("0");
```

```
    while (completed < n) {
        for (int i = 0; i < n; i++) {
            if (rt[i] > 0) {
                printf(" --P%d-- ", i + 1);

                if (rt[i] > tq) {
                    time += tq;
                    rt[i] -= tq;
                } else {
                    time += rt[i];
                    wt[i] = time - bt[i];    // WT = completion time - burst
time
                    rt[i] = 0;
                    completed++;
                }
                printf("%d", time);
            }
        }
    }

    printf("\n\nProcess\tBurst Time\tWaiting Time\tTurnaround
Time\n");
    for (int i = 0; i < n; i++) {
```

```
        tat[i] = bt[i] + wt[i];
        avg_wt += wt[i];
        avg_tat += tat[i];

        printf("P%d\t\t%d\t\t%d\t\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);

    return 0;
}
```

```
Enter number of processes: 5
Enter burst time of each process:
P1: 5
P2: 7
P3: 8
P4: 9
P5: 10
Enter time quantum: 4

Gantt Chart:
0 --P1-- 4 --P2-- 8 --P3-- 12 --P4-- 16 --P5-- 20 --P1-- 21 --P2-- 24 --P3-- 28 --P4-- 32 --P5-- 36 --P4-- 37 --P5-- 39

Process Burst Time      Waiting Time    Turnaround Time
P1              5               16              21
P2              7               17              24
P3              8               20              28
P4              9               28              37
P5              10              29              39

Average Waiting Time = 22.00
Average Turnaround Time = 29.80
```

PRIORITY

```
#include <stdio.h>

int main() {
    int n;
    int bt[20], wt[20], tat[20], pr[20], p[20];
    int i, j, pos, temp;
    float avg_wt = 0, avg_tat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority for each process:\n");
    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i+1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i+1);
        scanf("%d", &pr[i]);
        p[i] = i+1;  // process ID
    }
```

```
// Sorting processes by priority (lower number = higher priority)
    for (i = 0; i < n; i++) {
        pos = i;
        for (j = i+1; j < n; j++) {
            if (pr[j] < pr[pos]) {
                pos = j;
            }
        }
        // Swap priority
        temp = pr[i];
        pr[i] = pr[pos];
        pr[pos] = temp;

        // Swap burst time
        temp = bt[i];
        bt[i] = bt[pos];
        bt[pos] = temp;

        // Swap process ID
        temp = p[i];
        p[i] = p[pos];
        p[pos] = temp;
    }
```

```
// Waiting time for first process = 0
    wt[0] = 0;

    // Calculate waiting time
    for (i = 1; i < n; i++) {
        wt[i] = 0;
        for (j = 0; j < i; j++) {
            wt[i] += bt[j];
        }
        avg_wt += wt[i];
    }

    // Calculate turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = bt[i] + wt[i];
        avg_tat += tat[i];
    }

    // Print results
    printf("\nProcess\tBurst Time\tPriority\tWaiting Time\tTurnaround Time\n");
    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t\t%d\t\t%d\t\t%d\n", p[i], bt[i], pr[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat/n);

    return 0;
}
```

Number of processes: 3
Burst times: 10, 4, 6
Priorities: 2, 1, 3

```
Process   Burst Time   Priority   Waiting Time   Turnaround Time
P2        4            1          0              4
P1        10           2          4              14
P3        6            3          14             20

Average Waiting Time = 6.00
Average Turnaround Time = 12.67
```